

Evaluating LLMs under perturbation

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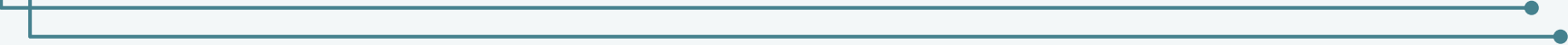
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What we found out?

04

Discussion

What it means for the field?



01.1

Introduction & Motivation

What are we doing



AI and Business



Growing Adoption

LLM agents enter business workflows.



High-Stakes Decisions

Outputs influence critical decisions.



Real-World Complexity

Operational data is imperfect.



Process Automation

Manual tasks become automated



Financial Applications

Used in risk assessments.



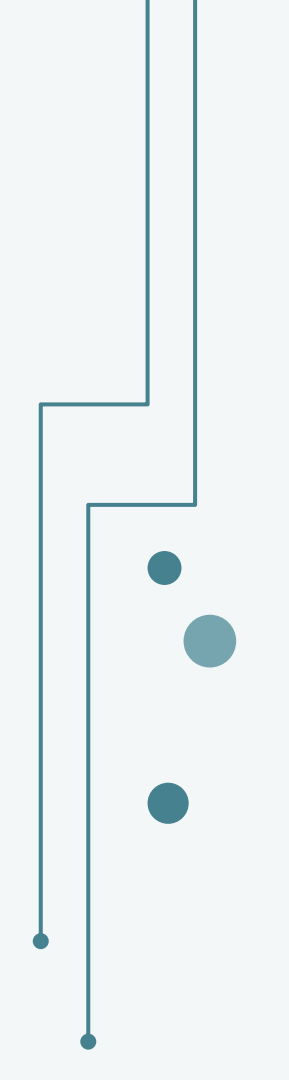
Deployment Readiness

Assess suitability before production



The Problem

Current benchmarks **evaluate agents under static and optimized conditions**, while real-world environments contain incomplete data, ambiguous inputs, changing requirements, and operational constraints. **As a result, benchmark success may not translate into reliable production performance.**



What does this mean for Business?



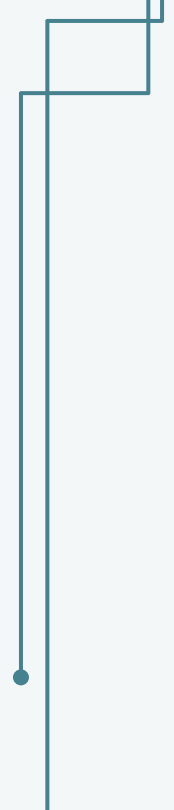
Unreliable

Organizations lack reliable methods to assess deployment readiness



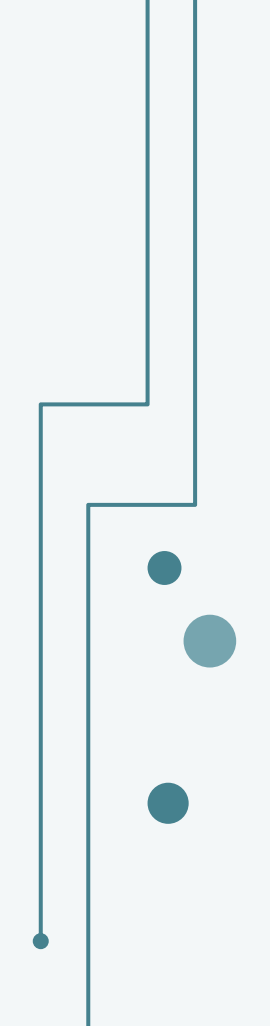
Magic Numbers

High benchmark performance does not guarantee operational reliability



The Motivation

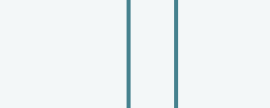
Despite strong benchmark performance, **there is limited evidence that existing evaluation methods accurately predict** how these systems behave under real-world operational conditions.





The Domain

The domain is characterized by complex reasoning requirements, evolving regulations, and **data that is often incomplete, inconsistent, or intentionally misleading**. Thus making it well suited for evaluating agent capabilities under realistic operational conditions.



02

Methodology

What were the steps?



02.1

LLMs

What we found out?



02.2

Agents

What we found out?

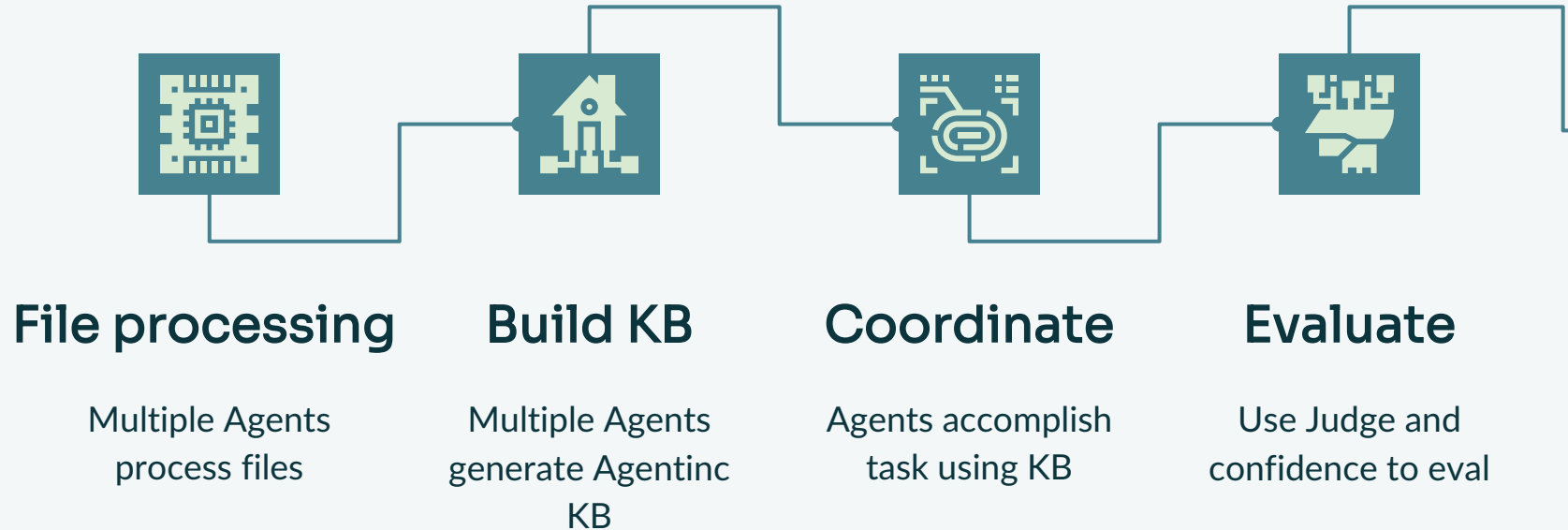


What are the metrics dimensions

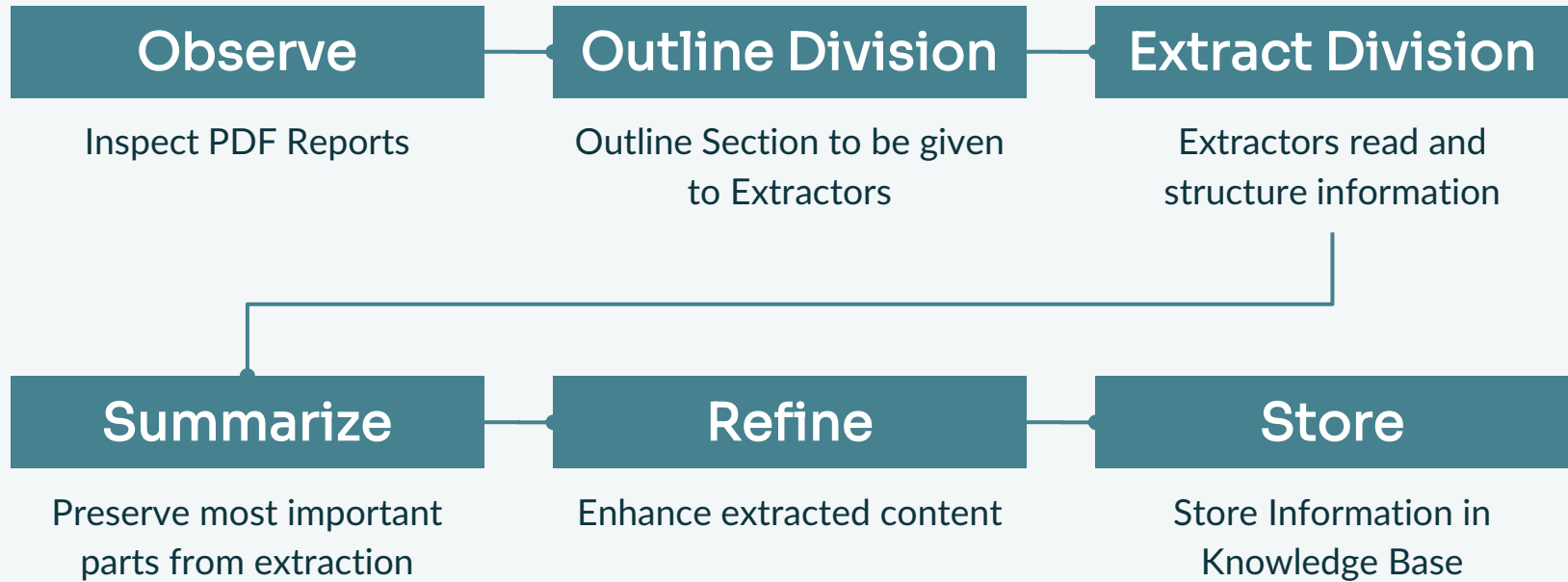
What do we wish to evaluate in terms of KPIs

<u>Reasoning</u>	How AI applies logic to evaluate information, infer conclusions, or break down problems
<u>Content Extraction</u>	Scanning and retrieving specific, relevant data points from structured or unstructured sources
<u>Agent Collaboration</u>	The interaction between agents that communicate, share resources, and divide labor
<u>Task Goal Completion</u>	The ability to successfully navigate all required steps, constraints, and edge cases
<u>Tool Usage and Context Understanding</u>	capacity to grasp the nuances of a user's prompt and recognize exactly when and how to invoke external functionalities
<u>Sub Agent Orchestration</u>	Coordinating and directing subordinate agents to complete parts of a larger workflow

Evaluating Agent Performance



File Processing



Build KB



Definition

Knowledge Base represents the cumulative isolated and filtered knowledge extracted from the reports

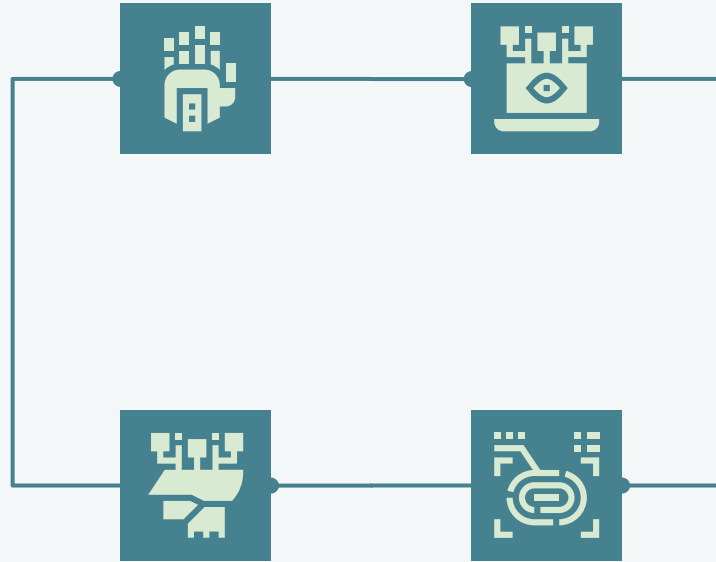


Advantages

Keeps data in uniform format and also easily accessible for Worker Agents. Provides the ability for context retrieval

Coordinate Agents

1. Spawn Manager Agents (MA)
2. MA orchestrate their own worker agents
3. Worker agents interact with the raw queries and retrieved information
4. MA formulated best answer and gives its confidence
5. MA provides its own Ground truth* and thinking



Judge Decides how
good an answer is based
on dataset ground truth
and Manager Agent
resources

Verifying answer



03

Results

What we found out?



03.1

LLMs

What we found out?



03.2

Agents

What we found out?



04

Discussion

What it means for the field?



DC

Limitations

What prevented us from doing better



Build KB



Time

Pipeline RUNs easily cross over 7 hours per run, thus making the process quite slow.



Available Data

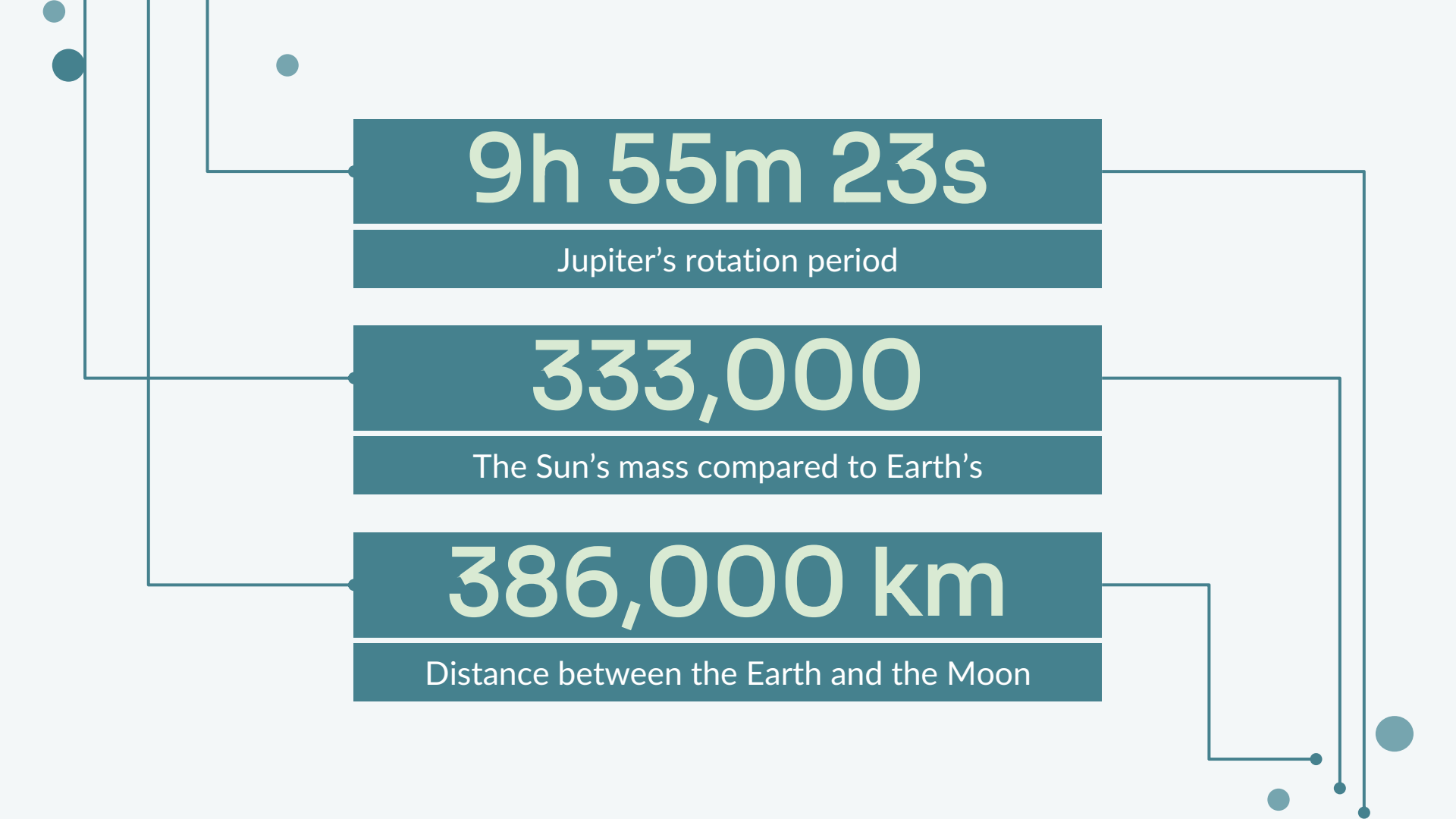
Only 2 Datasets and close to 200 PDFs were available on the topic.



Thanks!

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9h 55m 23s

The infographic features a light blue background with a network of thin, dark blue lines. On the left side, there are three solid dark blue circles of varying sizes. On the right side, there are several smaller dark blue circles and dots, some connected to the main lines. The three facts are presented in a vertical stack, each with a dark teal rectangular box for the value and a lighter teal box for the description. The lines connect the descriptions to their respective values.

Jupiter's rotation period

333,000

The Sun's mass compared to Earth's

386,000 km

Distance between the Earth and the Moon

Introduction

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god. Mercury's surface is filled with craters

Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there! Since the temperatures are so extreme, albeit not as extreme as on Venus, Mercury has been deemed to be non-habitable for humans

AI and ethics



Transparency

Mercury is the closest planet to the Sun and the smallest one of them all



Privacy

Venus has a beautiful name and is the second planet from the Sun



Legal

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

Examples



Business

Mercury is the closest planet to the Sun and the smallest one of them all



Entertainment

Venus has a beautiful name and is the second planet from the Sun



Social

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

AI and business



Applications

Despite being red,
Mars is very cold



Models

Venus is the second
planet from the Sun



Role of data

Neptune is the
farthest planet from
the Sun



Ethics

Mercury is the closest
planet to the Sun



Customer

Saturn is a gas giant
and has several rings



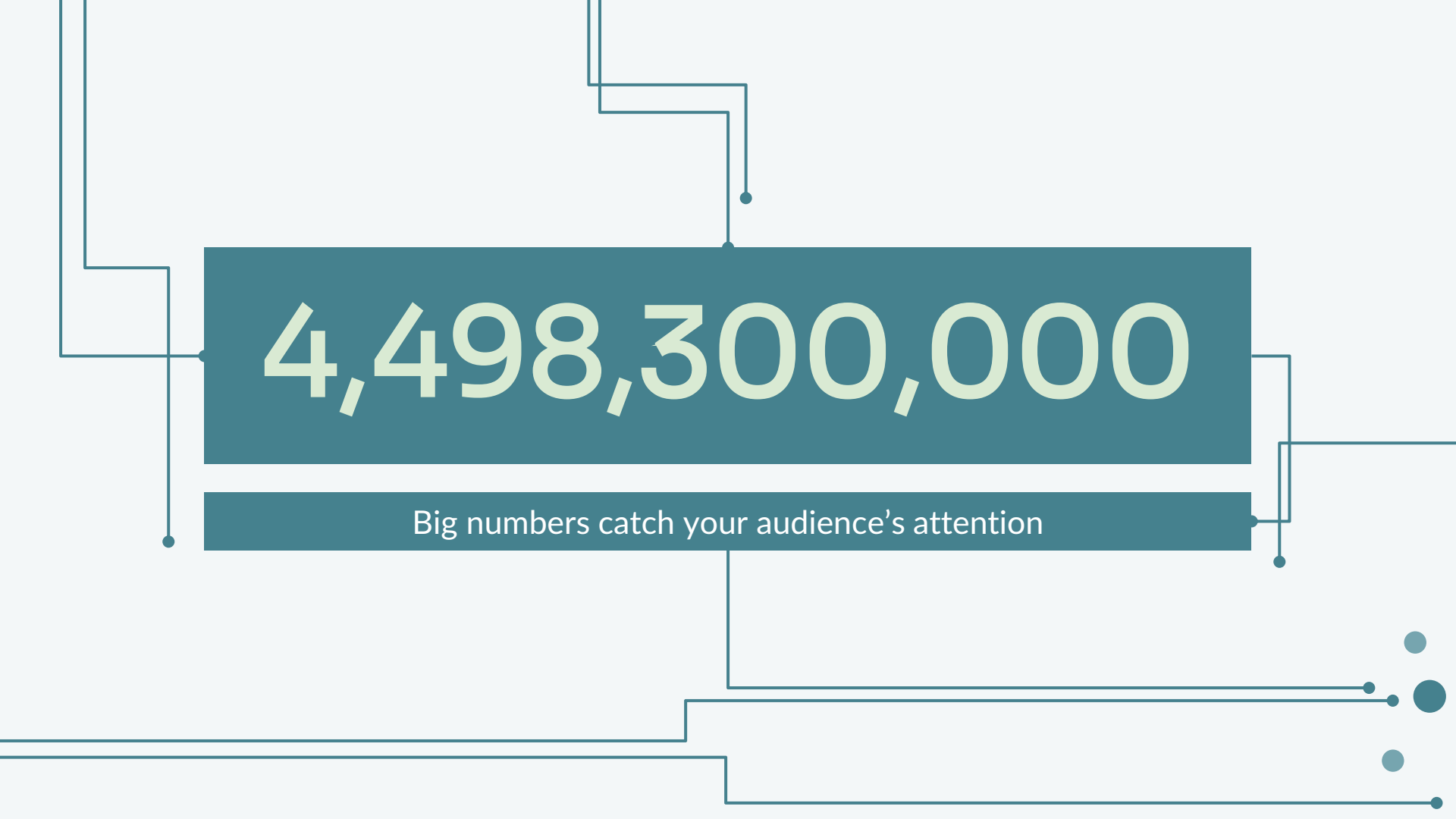
Future

Jupiter is the biggest
planet of them all



A picture always reinforces the concept

You can give a brief description of the topic you want to talk about here. For example, if you want to talk about Mercury, you can say that it's the smallest planet in the entire Solar System

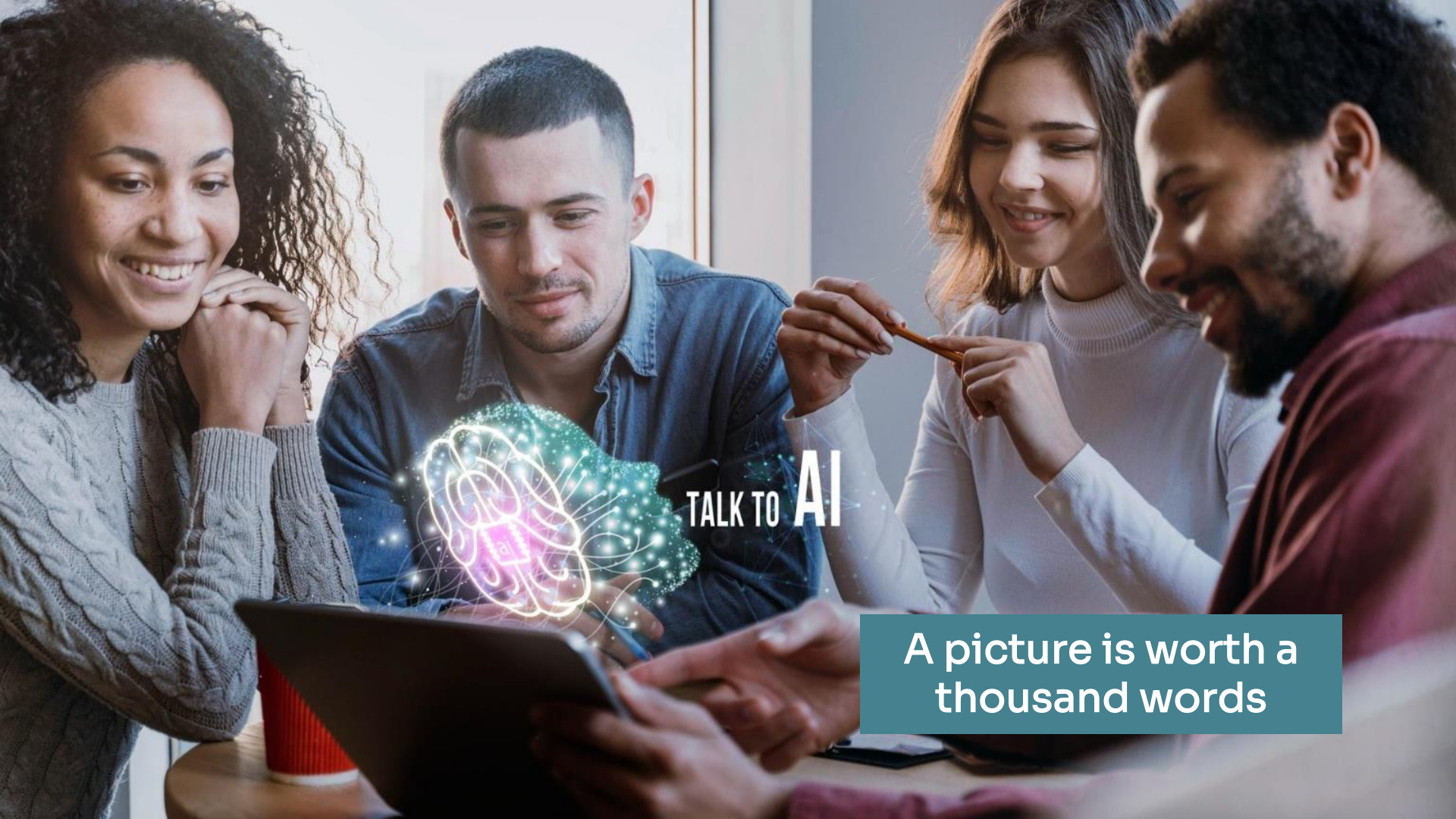
An abstract graphic featuring teal lines and dots. The lines are of varying thickness and form a network-like structure, with some lines ending in small teal dots. The background is a light gray. The main focus is a large teal rectangle containing the number 4,498,300,000 in a light green, sans-serif font. Below this rectangle is a smaller teal rectangle containing the text 'Big numbers catch your audience's attention' in a white, sans-serif font. The overall design is modern and minimalist.

4,498,300,000

Big numbers catch your audience's attention

Awesome words





TALK TO AI

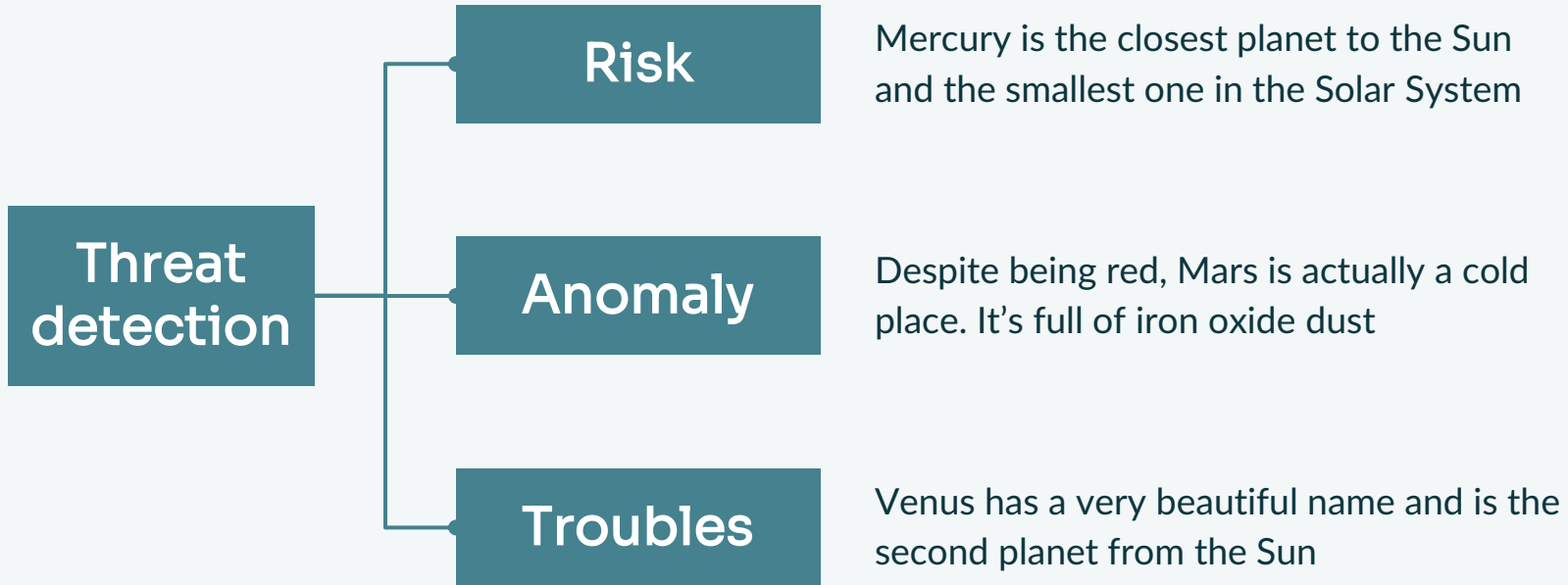
A picture is worth a
thousand words

KWL (brainstorming)

What I know	Mercury is the closest planet to the Sun
	Despite being red, Mars is a very cold planet
What I want to know	Venus is the second planet from the Sun
	Jupiter is the biggest planet of them all
What I learned	Saturn is composed of hydrogen and helium
	Neptune is the farthest planet from the Sun



Advanced threat detection



AI and transportation

Monitoring

Despite being red, Mars is actually a very cold place



Predictive

Neptune is the farthest planet from the Sun

Reduce waste

Mercury is the closest planet to the Sun

Advantages

Saturn is a gas giant and has several rings

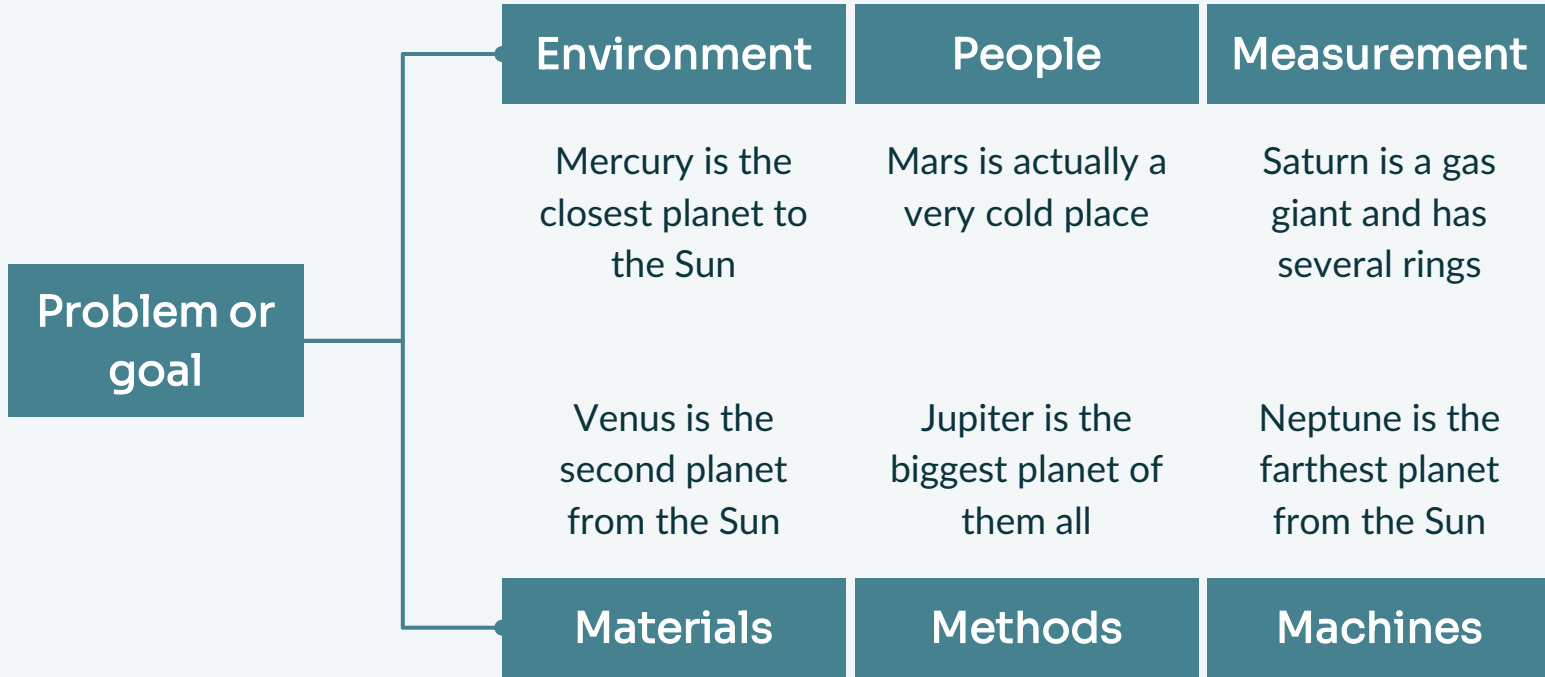
Future

Jupiter is the biggest planet of them all

Classification of AI

Reactive machines	Limited memory	Theory of mind	Self-aware conscious
● Small ● Red ● Cold ● Rocky	● Small ● Hot ● Dry ● Volcanic	● Small ● Hot ● Rocky ● Cratered	● Large ● Cold ● Gassy ● Striped
Despite being red, Mars is actually a cold place	Venus is extremely hot, even more than Mercury	Mercury is the smallest planet of them all	Jupiter is the biggest planet of them all

Cause and effect



Theory of mind



Why is important the theory of mind?

Is Mercury the closest planet to the Sun and the smallest one in the Solar System? Note that it's a bit larger than the Moon



Understand the mind of humans

Venus has a beautiful name and is the second planet from the Sun. It's hot and has a poisonous atmosphere

Parts and whole

The whole object

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal

Parts of the object

Mercury

Mars

Saturn

Venus

Jupiter

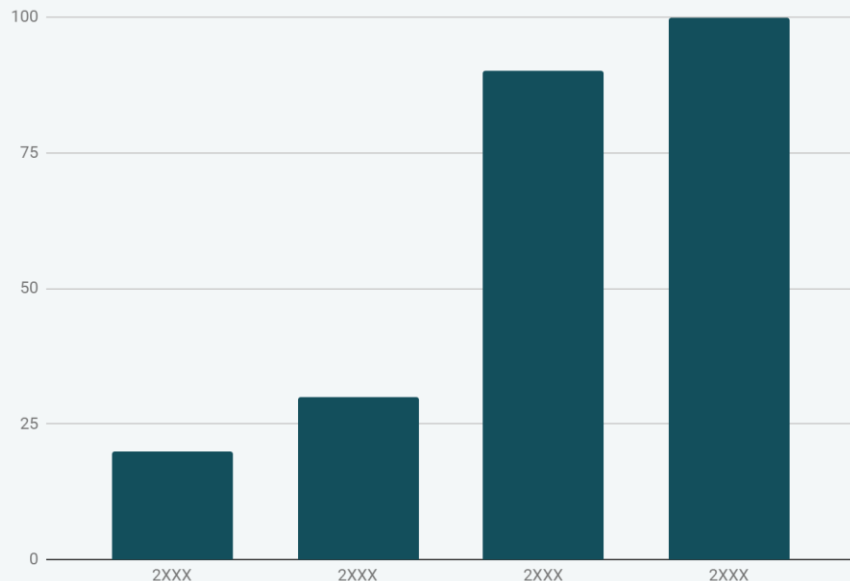
What happens if the parts are missing?

Venus has a beautiful name and is the second planet from the Sun

What's the function of the parts?

Saturn is a gas giant and has several rings. It's composed of hydrogen and helium

Evolution of domestic AI



Evolution

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. This planet's name has nothing to do with the liquid metal

Follow the link in the graph to modify its data and then paste the new one here. **[For more info, click here](#)**

Where is AI being studied



● Universities

Venus is the second planet from the Sun

● Institutions

Mercury is the first planet from the Sun

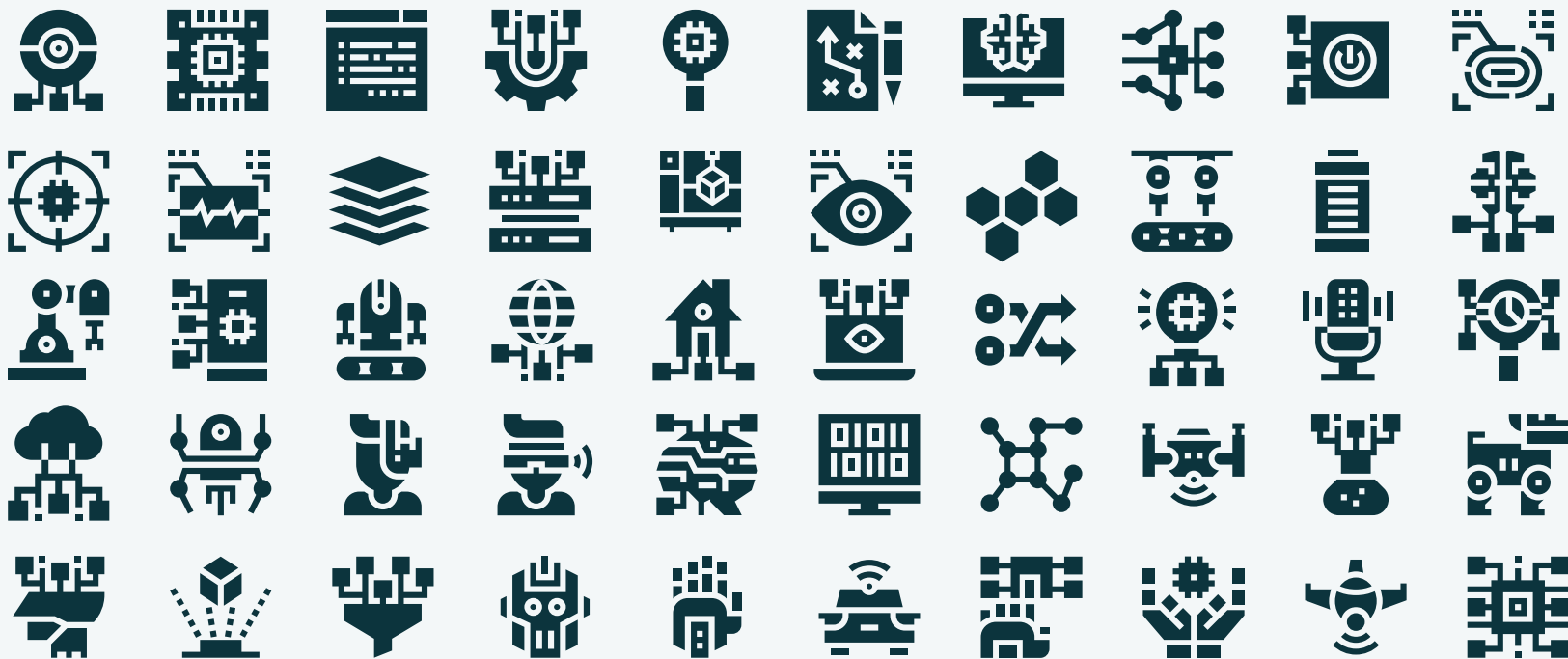


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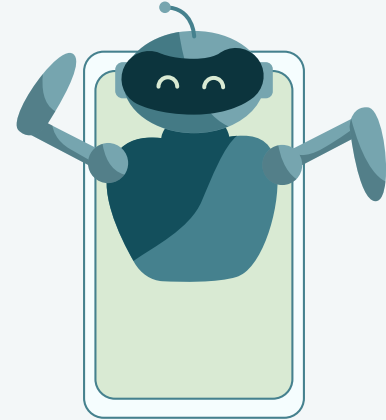
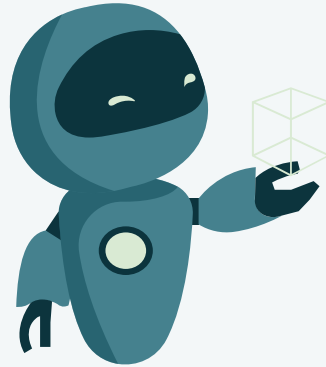
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Pana



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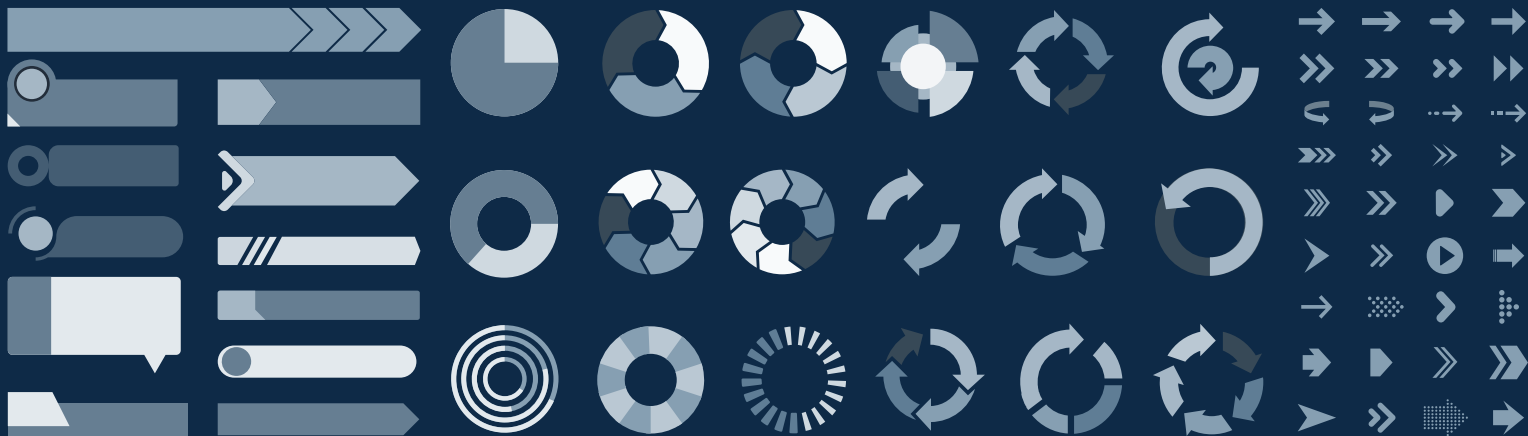
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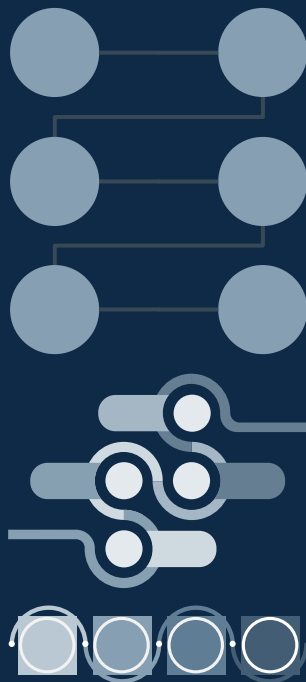
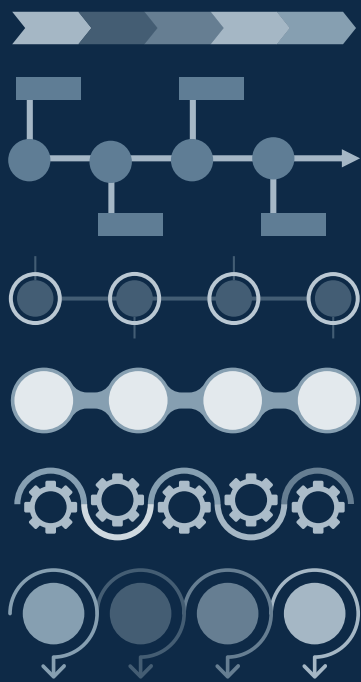
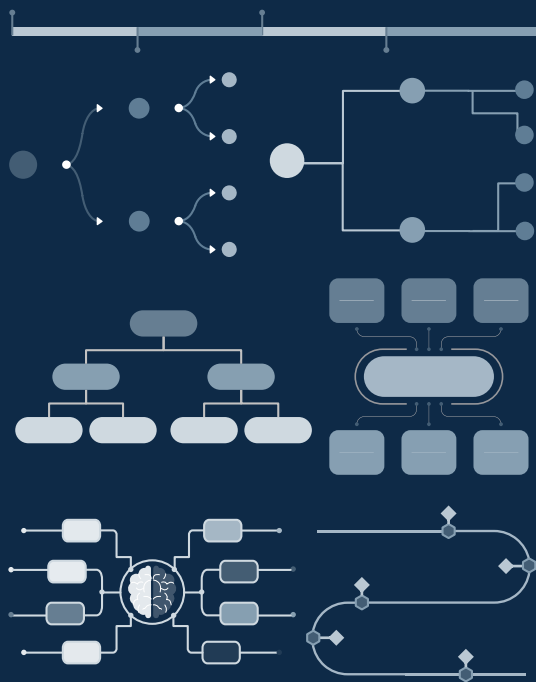
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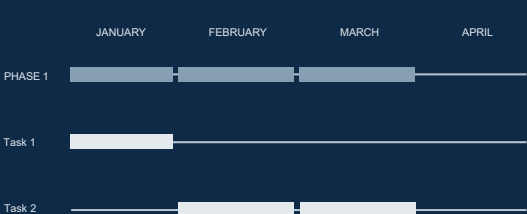
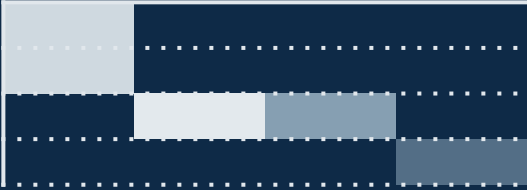
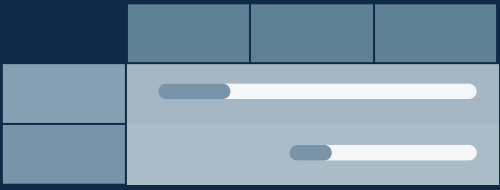
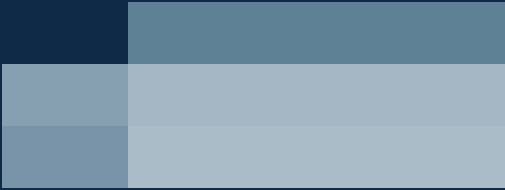
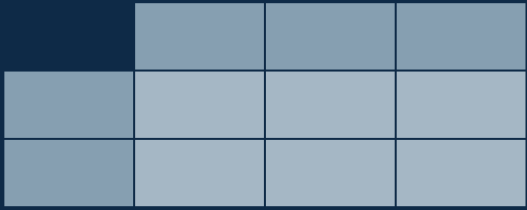
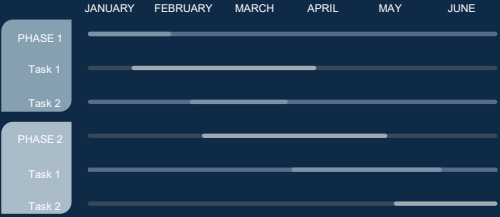
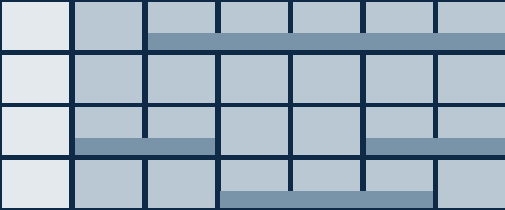
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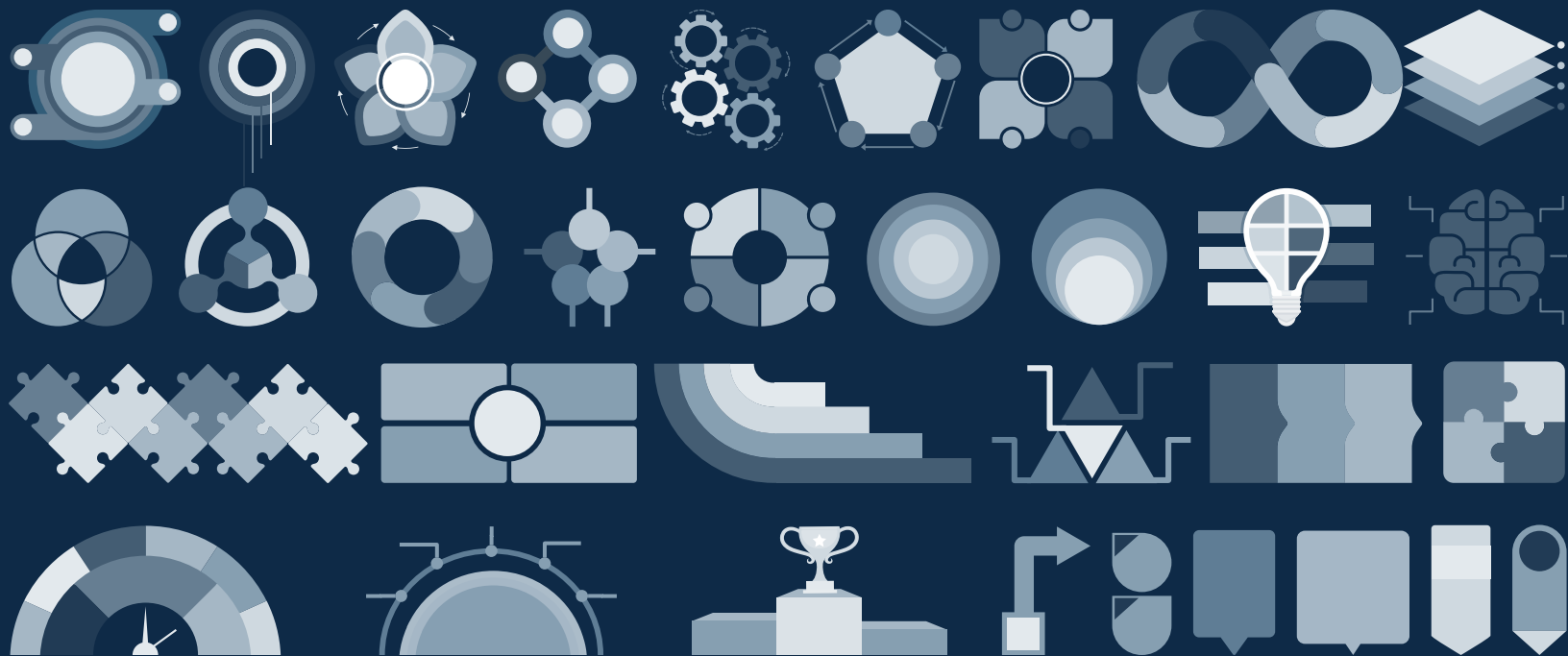
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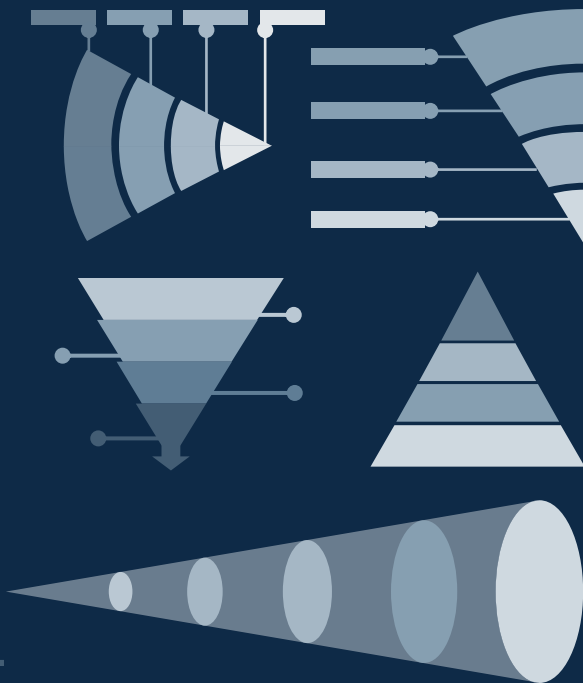
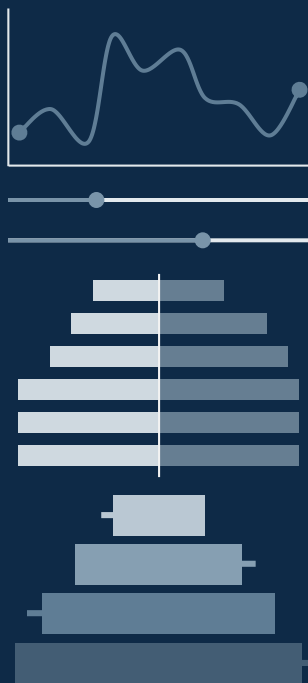
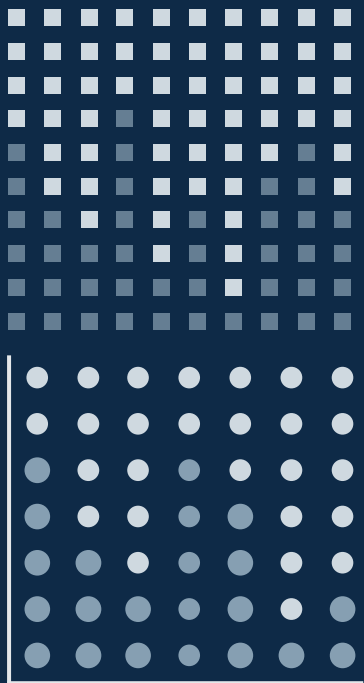












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